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TBA4510 - GEOTECHNICAL ENGINEERING, SPECIALIZATION
PROJECT

Finite Element Methods for Lower and Upper Bound Limit Analysis

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1 Introduction

Stability analysis is a fundamental aspect of geotechnical engineering and is essential for safe design of foundations, slopes, excavations, and retaining structure. Stability problems are governed by the ultimate capacity of the soil, where failure may occur once the applied loads exceed the soils shear strength. Incorrect prediction of collapse loads can lead to severe consequences, including excessive deformation or structural failure.Chen (1975)

Analytical solutions to stability problems are available only for high idealized cases involving simple geometries and homogeneous soil conditions. Classical solutions, such as the Prandtl bearing capacity mechanism for undrained soil, provide valuable theoretical insight but are rarely applicable to real engineering problems. As a result, numerical methods have become essential for modern geotechnical analysis.Chen (1975)

Among numerical approaches, the finite element method (FEM) is widely used due to its flexibility in handling complex geometries and material behavior. However, conventional FEM based strength reduction techniques, such as those implemented in PLAXIS, provide a single collapse estimate without explicitly indicating whether the solution is upper or lower bound to the true solution. Limit analysis offers an alternative framework by directly estimating collapse load thorough rigorous upper and lower bounds, providing a clear measure of solution reliability.

Finite element limit analysis (FELA) combines limit analysis theory with finite element discretizations, allowing complex geotechnical problems to be solved as optimization problems Sloan (1988). Recent advances in optimization techniques, particularly second order conic programming (SOCP) have significantly improved the efficiency and robustness of FELA Makrodimopoulos and Martin (2006). These developments are now implemented in software such as OPTUMG2, making advanced limit analysis accessible for practical engineering application Krabbenhøft et al. (2016).

1.1 Motivation

Designing safe and economical geotechnical structures requires reliable estimates of the collapse load and a clear understanding of the uncertainty associated with numerical prediction. While analytical solutions define exact collapse mechanism for idealized problems, their applicability is limited. Numerical method must therefor be carefully assessed to ensure they provide accurate and conservative resultsChen (1975).

A key challenge in numerical limit analysis is mesh dependency.The accuracy, efficiency, and robustness of FELA solution are strongly influenced by the mesh quality and refinement strategy. Uniform mesh refinement requires a substantial increase in element density to converge, while adaptive mesh refinement achieves comparable accuracy by locally refining regions in zones of plastic deformation, with out increasing element number through the entire model.

This project is motivated by the need to better understand how lower and upper bound FELA solution converges towards the exact collapse load, and how mesh refinement influence this convergence. Particular focus is placed in the implementing how lower bound FELA, as it provides conservative estimates that are specially valuable for geotechnical design.

By comparing analytical solutions with numerical results obtained using OPTUMG2, this study aims to assess the effectiveness of SOCP based FELA and adaptive meshing in reproducing the collapse mechanism and delivering reliable bearing capacity predictions. The findings contribute to improved confidence in limit analyses as a practical and robust tool for geotechnical stability assessment.

1.2 Research Objectives

Main objective

The main objective of this project is to investigate the application of finite element limit analysis for computing lower and upper bound solutions to a collapse load, and to evaluate how these bounds converge towards the exact analytical solution through numerical implementation and mesh refinement.

Specific objective Specifically, this research will address the following research questions

- To develop a solid theoretical understanding of limit analysis, including lower and upper bound theorem, and to analytically demonstrate how these bounds converge towards the exact collapse load.
- To introduce SOCP as a efficient numerical optimization framework used for solving finite element limit analysis problems.
- To apply an SOCP based FELA implementation model OPTUMG2 to compute lower and upper bound collapse loads for a selected case study.
- To investigate the effect of uniform and adaptive mesh refinement, and to assess how mesh refinement influence the convergence and numerical accuracy of the bounds.

1.3 Summary of Chapters

Chapter 1 presents the motivation for the study, the research objectives, and an overview of the structure of the report.

Chapter 2 provides the theoretical background for the project, including Mohr Coulomb failure criterion, idealized models and the fundamentals of limit analysis.

Chapter 3 presents a literature review, stability analyses methods, and the development of LB FELA and implementation of lower bound limit theory.

Chapter 4 applies the theoretical framework to a practical bearing capacity problem. Analytical UB and LB solutions are derived and compared with numerical results from FELA. The effects of mesh refinement and the convergence of the collapse load are also examined.

Chapter 5 summarises the key findings of the study, outlines the strengths and limitations of the methods used, and recommends areas for further research, particularly improved mesh refinement strategies and deeper exploration of SOCP based lower bound FELA.

2 Background

This chapter presents the theoretical background necessary for the analysis conducted in this project. It introduces the fundamental concept of stress representation in soils, yield criteria, and idealized soil behavior, which form the basis for stability analysis in geotechnical engineering. The principles of limit analysis, including the lower and upper bound theorem, are then introduced, as they form the theoretical framework for both the analytical solutions and the numerical methods employed later in this project.

2.1 Characterization of Stress states

In geotechnical engineering, describing the stress conditions within the soil is fundamental, as soil strength and deformation depend strongly on the applied stresses. At any point within a soil mass, the stress state can be represented by the normal and shear stress acting on the faces of an infinitesimal element as shown in figure 1 (Timoshenko & Goodier, 1970, p. 2-7).

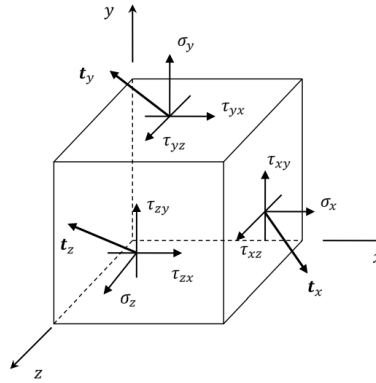


Figure 1: Stress component acting on an infinitesimal element. Adapted from Timoshenko and Goodier (1970, p. 4)

These stress components can be assembled into a second order stress tensor, which provides a complete mathematical description of stress state at a point: (Timoshenko & Goodier, 1970, p. 2-7)

$$\sigma_{ij} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} \quad (1)$$

2.1.1 Symmetry of the the stress tensor

Figure 2 illustrates moment equilibrium of an infinitesimal element. To satisfy rotational equilibrium, opposing shear stresses must be equal. This requirement leads to symmetry of the stress tensor: (Timoshenko & Goodier, 1970, p. 5)

$$\tau_{xy} = \tau_{yx} \quad \tau_{yz} = \tau_{zy} \quad \tau_{xz} = \tau_{zx}$$

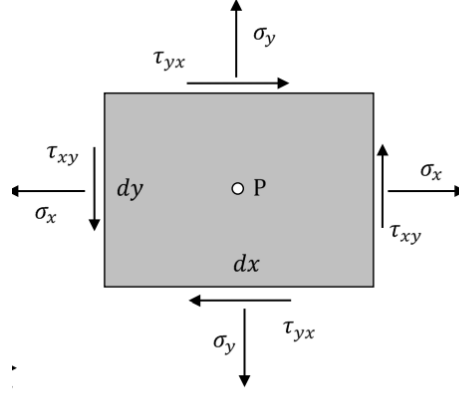


Figure 2: Moment equilibrium of a plane element illustrating shear stress symmetry. Adapted from Timoshenko and Goodier (1970, p. 5)

2.1.2 Plane strain condition

Through out this project, plane strain conditions are assumed, meaning that deformation in the out of plane direction is neglected ($\epsilon_z=0$). As a result, the three dimensional stress state reduces to a two dimensional representation, simplifying the analysis.(Timoshenko & Goodier, 1970, p. 5)

$$\sigma_{ij} = \begin{bmatrix} \sigma_x & \tau_{xy} \\ \tau_{yx} & \sigma_y \end{bmatrix} \quad (2)$$

Due to shear stress symmetry $\tau_{xy} = \tau_{yx}$, the stress state may be presented by the stress vector

$$\sigma = \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} \quad (3)$$

2.1.3 The Mohr Coulomb yield criterion

The Mohr Coulomb yield criterion is one of the most widely used failure models in geotechnical engineering for frictional soils. It describes the condition under which soil begins to yield due to shear failure. The stress state at a point can be represented using a Mohr circle, which provides a simple way to visualize the relationship between normal and shear stress acting on the horizontal and vertical planes of a soil element.(Norwegian University of Science and Technology, 2023, p. 129-131)

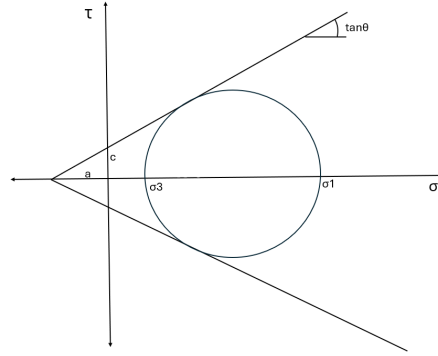
In the Mohr Coulomb model, failure occurs when the Mohr circle becomes tangent to a linear failure envelop defined by the soil strength parameters: cohesion c and friction angle ϕ . The envelope represents the dependency of shear strength on normal stress and is expressed as (Norwegian University of Science and Technology, 2023, p. 129-131)

$$\tau_f = c + \sigma \tan \phi$$

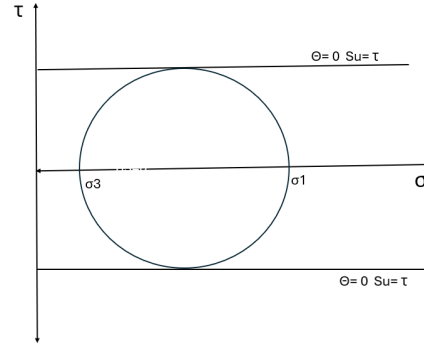
As shown in figure 3a the inclination of the envelope reflects the frictional resistance of the soil, while the intercept represents its cohesive strength.

When the friction angle is zero ($\phi = 0$), the Mohr Coulomb criterion reduces to a special case in which shear strength is independent of normal stress and governed solely by the undrained shear

strength S_u . This condition is typical of undrained loading in a saturated clays, where excess pore water pressure prevents the mobilization of the frictional resistance. The resulting yield behavior is described by the Tresca criterion, which can be regarded as a special case of the Mohr Coulomb model. (Norwegian University of Science and Technology, 2023, p. 134-135)



(a) Illustration of the Mohor-Coulomb criterion



(b) Illustration of the Tresca criterion

2.1.4 The Mohr Coulomb yield surface

When the Mohr Coulomb criterion is expressed in terms of the three principal stresses, it no longer appears as a simple line but instead forms a yield surface in stress spaces. This surface defines the boundary between elastic behavior and plastic yielding of the soil. In three dimensional principal stress space, the Mohr Coulomb yield surface takes the shape of a hexagonal pyramid cone, indicating that yielding depends on the relative magnitudes of the principal stresses. (Chen, 1975, p. 17-21)

This hexagonal geometry represents the exact, non smooth geometry of the criterion. For numerical application, this surface is often approximated or formulated in alternative representations, depending on the optimization framework being used. (Chen, 1975, p. 17-21)

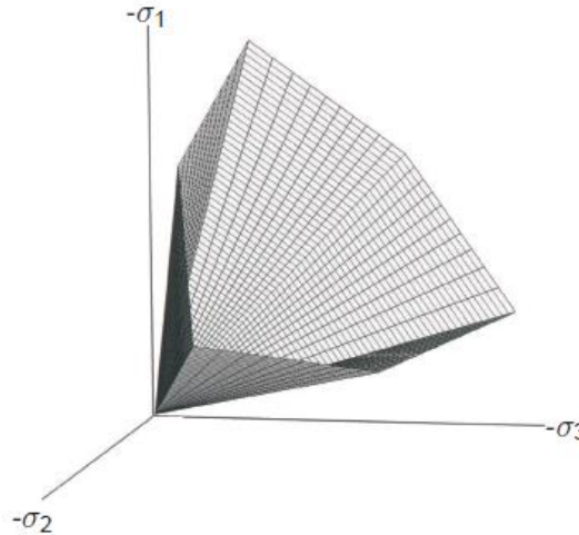


Figure 4: Principal stress space representation of Mohr Coulomb criterion. Adapted from Yilmaz (2019, p. 9)

2.2 Idealized soil model engineering analysis

In geotechnical engineering, foundations must safely transfer structural loads to the ground while limiting settlement and preventing collapse. Understanding soil behavior under loading is therefore essential for analyzing both deformation and failure. The true stress- strain behavior of soil is highly nonlinear, involving yielding, permanent deformation, and often strain softening.(Chen, 1975, p. 15-17)

To make engineering analysis practical, this behavior is commonly idealized using the linear elastic perfect plastic model. In this representation, the soil responds elastically up to a yielding point and then deforms plastically at constant stress, without hardening or softening until failure occurs, as shown in figure 5 (Chen, 1975, p. 15-17)

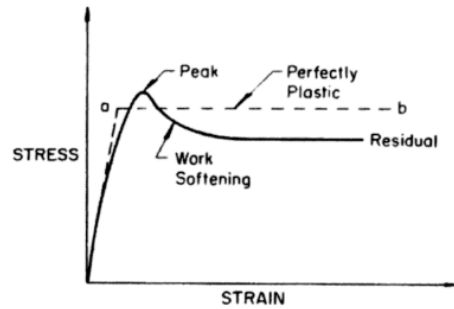


Fig. 2.1. Stress-strain relationship for ideal and real soils.

Figure 5: Comparison between idealised elastic-perfectly plastic and real soil stress-strain behaviour. Adapted from Chen (1975, p. 16)

2.3 Basic principles of the limit analysis

Limit analysis is widely used in geotechnical engineering to estimate collapse loads and assess stability. It is commonly applied to problems such as bearing capacity, slope stability, retaining walls, and other situations where soil failure may occur. The method is based on two principles, the upper bound principle and the lower bound principle(Chen, 1975, p. 15).

2.3.1 Lower bound principle

The lower bound solution is obtained by considering a statically admissible stress fieldSloan (2005).

A stress field is considered statically admissible if it satisfies the following conditions:

- equilibrium within the soil domain
- the stress boundary conditions, and
- does not violate the yield criterion

2.3.2 Upper bound principle

The upper bound principle estimates load by assuming kinematically admissible velocity fieldSloan (2005). An admissible upper bound mechanism must satisfy:

- the strain - displacement relations,

-
- the associated flow rule, and
 - the displacement boundary conditions

The resulting estimate is an upper bound because the ratio of external work to internal plastic dissipation is always greater than or equal to the true collapse load.(Norwegian University of Science and Technology, 2023, p. 137)

3 Literature review

3.1 Stability analysis methods

Stability analysis methods are used in geotechnical engineering to assess whether a soil mass can sustain applied loads without collapse. In addition to limit analysis, geotechnical stability problems are commonly analyzed using other methods, that can be classified as the: the limit equilibrium method and the slip line methods (Chen, 1975, p. 17). All this method will be discussed in this section.

3.1.1 Limit equilibrium method

The limit equilibrium method is the earliest method used for stability analysis, and was first applied in geotechnical engineering by Coulomb (1773). This method has been widely used to obtain approximate solutions for stability problem by assuming potential failure surfaces. To estimate the limit load, a possible failure shape is assumed, and the stress acting along this failure surface are governed by the soil strength parameters, namely cohesion c and friction angle ϕ (Sloan, 2005, p. 1; Chen, 1975, p. 22).

The application of this method have been developed further by Terzaghi (1943), and Taylor (1948), (Chen, 1975, p. 22). The main advantages of this method are its simplicity and widespread application over a long time, which have resulted in the availability of well-established computational tools that can solve stability problems. (Sloan, 2013, p. 1; Chen, 1975, p. 22)

However, this method is limited by the need to predefine the geometry of the failure surface, an incorrect assumption may result in inaccurate estimates of the failure load. Under realistic conditions, particularly in cases involving irregular geometries, complex loading, or layered soil profiles the actual shape of failure is often difficult to define in advance. Further challenges include accounting for anisotropic and inhomogeneous materials and extending the method to three dimensions. (Sloan, 2005; Chen, 1975, p. 22)

Despite these limitations, numerous limit equilibrium formulations have been proposed and successfully applied in the field of geotechnical engineering. Sloan (2013)

3.1.2 Slip-line method

The slip-line method is theoretical approach used to describe soil plasticity and the stress state that develop when soil reaches a fully plastic condition University of Cambridge (2004/2025). This method is based on plasticity theory and is commonly applied to wide range of geotechnical problems. (Chen, 1975, p. 20-22)

When applying the slip-line method, the soil mass is assumed to behave as an ideal rigid-perfectly plastic material. As discussed earlier, stability analysis focuses solely on the soil at the failure zone. Soil yielding is assumed to follow the Mohr-Coulomb failure criterion, which provides a simple and widely accepted representation of shear strength. Its linear form makes it particularly suitable for slip-line analysis. In addition, the plastic zone is assumed to deform freely once yielding occurs. Under this assumption, no additional constraints limits deformation within the plastic region, allowing the stress state to adjust naturally along slip-lines, which is essential for identifying the failure mechanism. Shin et al. (2023)

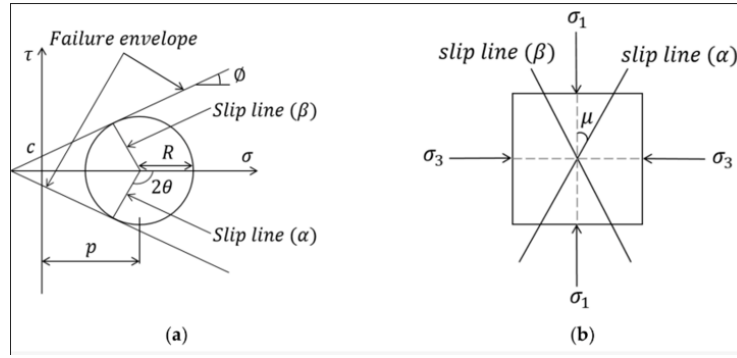


Figure 6: Mohr circle representation of the plastic stress state in a Coulomb material, (a) Slip-lines represented using Mohr circle notation (b) Stress state at a point within a soil mass. Adapted from Shin et al. (2023)

Under plane strain conditions, the principle stress at each point act in perpendicular direction. By connecting these stress directions from point to point, two smooth curves can be formed. These curves represent the paths along which stress is transmitted through the material, and are known as principal stress trajectories. At each point, there are also two preferred directions along which failure is most likely to occur, as illustrated 6b. When these failure directions are connected throughout the material, two continuous curves are obtained, referred to as slip lines. Shin et al. (2023)

For a slip line field to present a valid solution, the stress distribution throughout the entire body mass, must satisfy equilibrium conditions and must not violet the yield criterion, including in regions outside the slip line field University of Cambridge (2004/2025).

Based on this concept, the slip-line method was developed as an analytical technique for estimating collapse loads under plane strain conditions by analyzing stress states in fully plastic soil and constructing slip-line fields that represent directions of maximum shear stress at failure (Chen, 1975, p. 21).

The development of slip line method can be traced to the work of Kötter (1903) , who is widely regarded as the fist to drive the governing slip line field equations for plastic deformation. A significant advancement was later achieved by Prandtl (1920), who obtained a close form analytical solution to these equations for the bearing capacity of a footing resting on weightless soil. This solution introduced the concept of a singular point from which a fan shaped pattern of straight is created. These fundamental ideas were later applied and further developed by Reissner (1924) and Novotortsev (1938). (Chen, 1975, p. 20)

In practical geotechnical applications, complex boundary conditions significantly increase the mathematical complexity of the slip line equation, making analytical solutions difficult to obtain. To address this challenge, several approximation solution approaches have been developed. Early on Sokolovski in 1960, introduced a numerical method based on finite difference approximations for solving the slip line equations. This approach enabled the systematic construction of slip line fields for geotechnical problems involving foundations, slopes, and retaining walls (Chen, 1975, p. 21).

While slip-line method produce realistic failure mechanisms, it generally overestimate the collapse loads. A further limitation is that it neglects deformation behavior out side the plastic zones.University of Cambridge (2004/2025) Although the slip line provides useful insight into stress distributions and failure mechanisms under idealized conditions, it also has important limitations. In particular, the method does not explicitly account for the detailed stress-strain behavior of soils. As a result, slip line analysis is mainly applicable to ultimate limit states, and not suitable for predicting deformations prior to failure. Despite these limitations, the slip line method remains a valuable theoretical tool in geotechnical engineering.(Chen, 1975, p. 20-21)

3.1.3 Principle of limit analysis

"Limit analysis is based on the plastic bounding theorems developed by Drucker et al. (1951), and assumes small deformations, a perfectly plastic material, and an associated flow rule" (Sloan, 2005, p. 3)

The principal of limit analysis is based on two fundamental theorems: the lower bound theorem and the upper bound theorem. The lower bound theorem provides a conservative estimate of the true collapse load by requiring the stress field to be statically admissible, while the upper bound theorem yields an estimate by assuming a kinematically admissible velocity field. Together, these theorems define bounds within which the true collapse load must exist (Sloan, 2005; Chen, 1975, p. 23).

3.2 Improvement and convergence of lower and upper bound

The exact collapse load lies between the lower and upper bound solutions. In this section, analytical formulation of both bounds are presented and progressively refined. By systematically improving the assumed stress fields and failure mechanisms, it is shown that the lower and upper bound solution converges toward exact collapse load. This analytical process demonstrates, how limit analysis bounds and approaches the true solution.

3.2.1 Refinement of lower bound solution

Lower boundary construction mechanism

Constructing and improving of the lower bound is based Sloan (2005)

A statically admissible lower bound stress field is constructed by:

1. dividing the soil mass into regions with uniform stress state separated by stress discontinuities,
2. enforcing equilibrium across each discontinuity, and
3. ensuring that all stresses lie on or within the yield criterion.

to illustrate the procedure, the soil beneath the footing is first divided into three uniform stress zones, as shown in figure 7. Zone 1 and 3 are symmetric and satisfy the free surface condition $\sigma_3 = 0$, zone 2 carries the applied load q .

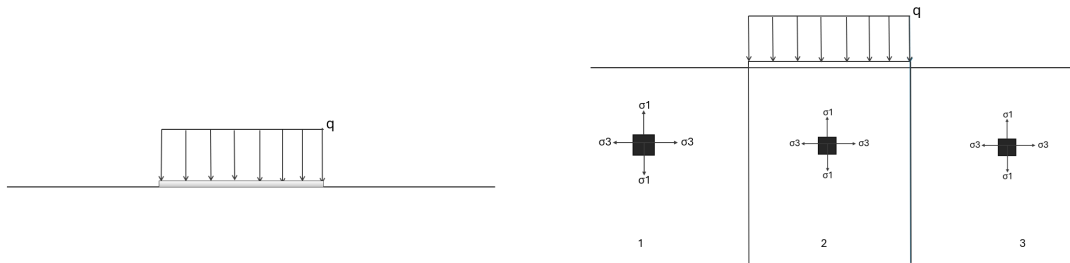


Figure 7: Three zone lower bound stress field beneath a footing

Three zone lower bound

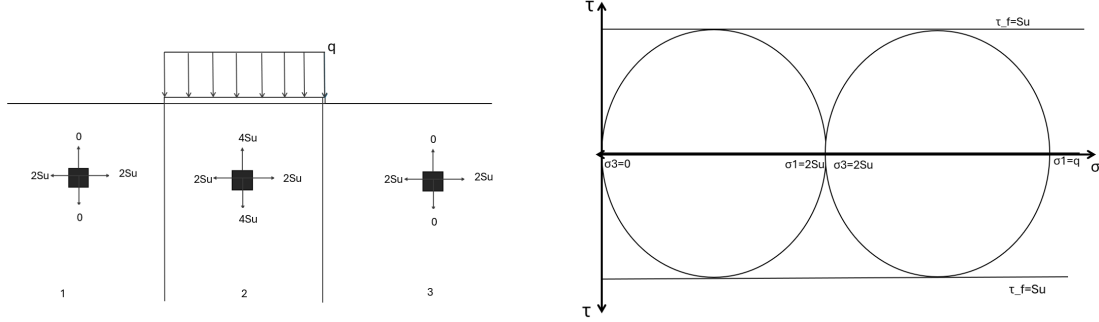
For a Tresca material, the yield surface is represented by Mohr circle with a diameter $2\tau_c$, illustrated in figure 8b. The stress in zone 2 becomes:

$$\sigma_3 = q - 2\tau_c$$

Enforcing equilibrium across the interfaces between zone 1, 2, and 3 gives:

$$q - 2\tau_c = 2\tau_c \Rightarrow q = 4\tau_c$$

This provides a basic conservative lower bound estimate, as the assumed stress field is relatively coarse.



(a) Three zone lower bound stress field, lower bound (b) Mohr circle representation of the three zone solution for S_u material lower state

Improved lower bound

A more accurate lower bound is obtained by introducing two additional stress zones separated by discontinuity planes, as shown in figure 9b. The stress path now follows:

$$\sigma_3 \rightarrow \sigma_A \rightarrow \sigma_B \rightarrow q$$

The corresponding stresses are:

$$\begin{aligned} \sigma_3 &= 0, \\ \sigma_A &= \tau_c + \tau_c \cos(\pi - 2\alpha), \\ \sigma_B &= \sigma_A + 2\tau_c \cos\left(2\alpha - \frac{\pi}{2}\right), \\ q &= \sigma_B + \tau_c \cos(\pi - 2\alpha) + \tau_c. \end{aligned}$$

Substitution gives:

$$q = 2\tau_c + 2\tau_c \left[\cos(\pi - 2\alpha) + \cos\left(2\alpha - \frac{\pi}{2}\right) \right]$$

Using trigonometric identities:

$$\cos(\pi - 2\alpha) = -\cos(2\alpha), \quad \cos\left(2\alpha - \frac{\pi}{2}\right) = \sin(2\alpha),$$

the expression simplifies to:

$$q = 2\tau_c + 2\tau_c [-\cos(2\alpha) + \sin(2\alpha)]$$

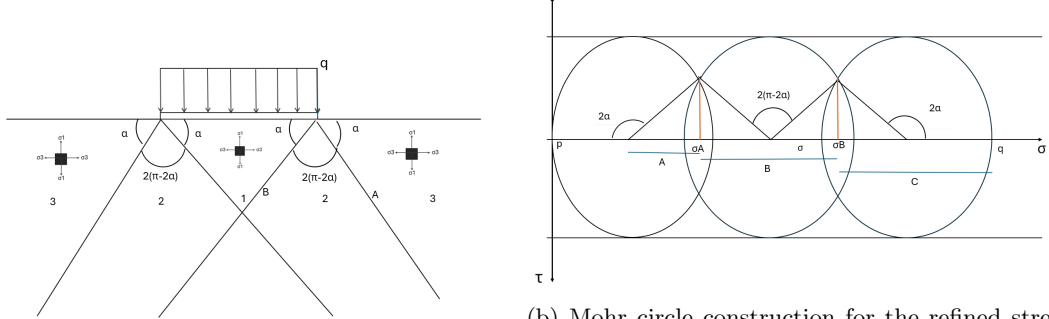
Differentiating with respect to α and setting the result to zero yields:

$$\tan(2\alpha) = -1 \Rightarrow \alpha = 67.5^\circ$$

Substituting back gives the improved the improved lower bound:

$$q = 2\tau_c + 2\sqrt{2}\tau_c = 4.828\tau_c.$$

Thus, by introducing additional stress discontinuities, the lower bound increased from $4\tau_c$ to $4.828\tau_c$.



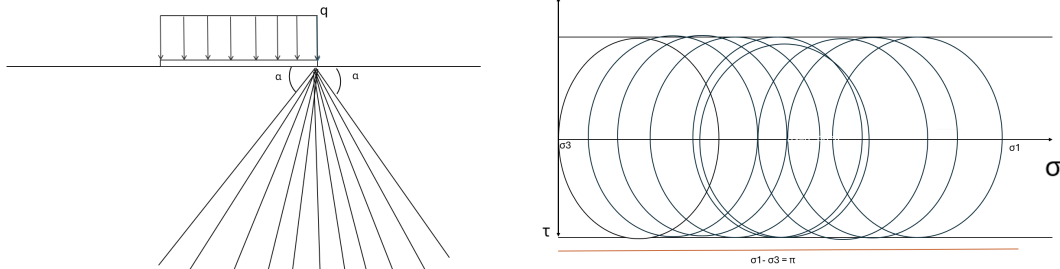
(a) Refine lower bound with additional stress zones field

Convergence toward the exact solution

As the number of zones increases, the lower bound solution continues to improve. Each zone corresponds to a point on the Tresca yield criterion. With increasing refinement, stress jumps smaller and more frequent (figure 10a, and the stress path approaches a smooth curve.

As the number of stress zones increases and tends to infinity, the stress path becomes continuous as shown in figure 11a, and the angular distance between σ_1 and σ_3 approaches π . Under these conditions, the lower bound solution converges to the exact collapse load for Tresca model:

$$q_{\text{critical}} = (2 + \pi)\tau_c \approx 5.14\tau_c$$



(a) Stress jumps for increasing number of zones approaching the Prandtl mechanism

(b) Continuous stress path yielding the exact solution $q = (2 + \pi) * \tau_c$

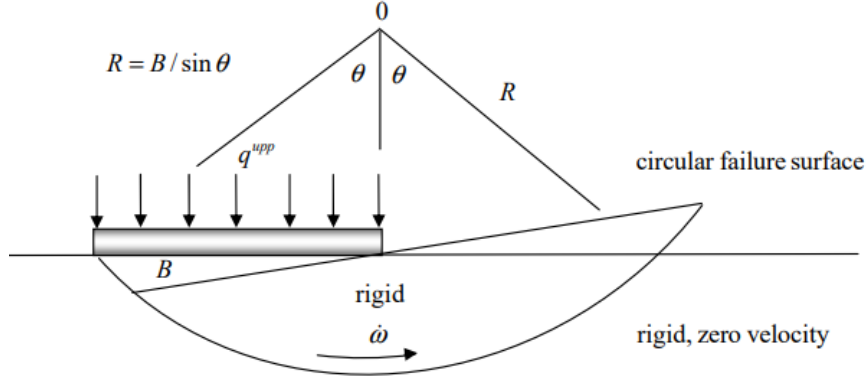
3.2.2 Refinement of upper bound solution

Constructing upper bound mechanism

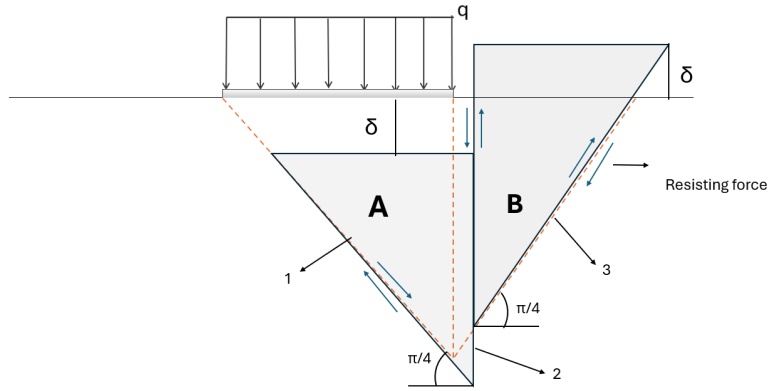
Constructing and improving of the upper bound is based Sloan (2005).

When a vertical surface load q is applied over a strip footing with width B under undrained conditions with constant shear strength τ_c , the soil may deform into a system of sliding blocks, shown in figure 11a. Due to symmetry, only one half of the mechanism is shown Sloan (2005).

As illustrated in figure 11bb, the soil mass is divided into rigid wedges separated by slip planes inclined at 45° . Block A moves downward and inward, while block B moves upward and outward.



(a) Upper bound bearing capacity mechanism beneath a footing. Adapted from (Sloan, 2005, p. 4)



(b) Sliding block upper bound mechanism with inclined plane at 45°

Internal work Internal energy dissipation occurs along the three slip labeled 1,2 and 3. The plastic dissipation along each plane is given by:

$$\tau_c \times (\text{sliding displacement}) \times (\text{length of the slip plane})$$

For slip plane inclined at 45°, the relative displacement along each plane is $\sqrt{2}\delta$. The internal dissipation along each plane is therefore:

$$\begin{aligned} W_1 &= \tau_c \left(\frac{B}{2} \sqrt{2} \right) (\delta \sqrt{2}), \\ W_2 &= \tau_c \left(\frac{B}{2} \right) (2\delta), \\ W_3 &= \tau_c \left(\frac{B}{2} \sqrt{2} \right) (\delta \sqrt{2}). \end{aligned}$$

Summing these contributions gives the total internal work:

$$W_{\text{int}} = 3 \tau_c B \delta$$

External work The external work is preformed by the applied load q acting through the vertical displacement δ :

$$W_{\text{ext}} = q \frac{B}{2} \delta$$

Collapse load from the upper bound Applying the upper bound theorem by equating external work and internal dissipation

$$W_{\text{ext}} = W_{\text{int}}$$

gives:

$$q = 6\tau_c,$$

The value represents an upper bound estimate of the true collapse load. Since the lower bound obtained earlier was $4\tau_c$, the actual collapse load must lie between these values.

Improved upper bound

The upper bound estimate can be refined by introducing additional sliding blocks between wedges A and B. As the number of blocks increase, the slip planes rotate and the mechanism gradually approaches the exact solution.

In the limit of infinitely many blocks, the upper bound solution converges to the exact collapse load for a Tresca material:

$$q_{\text{exact}} = (2 + \pi)\tau_c, \approx 5.14 \tau_c.$$

3.2.3 Exact solution

Both the lower and upper bound solutions can be systematically improved by refining the assumed stress field. In the lower bound approach, improvement is achieved by increasing the number of stress zones and reducing the size of stress discontinuities. As more zones are introduced, the stress path along the yield surface becomes smoother and the estimated collapse load increase while remaining lower than the true solution Sloan (2005).

Similarly, the upper bound solution can be improved by adopting more detailed kinematically admissible failure mechanisms. Introducing additional sliding blocks or allowing continuous velocity fields reduces the calculated external work relative to the internal dissipation, causing the upper bound estimate to decrease Sloan (2005).

As the level of refinement increase, the lower bound solution increases and the upper bound solution decreases, leading to convergence toward a common value. For undrained Tresca material this value is

$$q_{\text{exact}} = (2 + \pi) \tau_c + p.$$

where $N_c = 2 + \pi$ is the bearing capacity factor, S_u is the undrained shear strength, and P is the overburden pressure.

As shown in figure 12 the collapse load mechanism for the exact solution matches very well with the Prandtl failure mechanism. The collapse mechanism originally proposed by Prandtl (1920) is characterized by three distinct regions beneath a footing. An active zone forms directly under the footing, where the soil moves downward under applied load. On either side of this region, passive zones develop in which soil is displaced upward and outward, mobilising resistance. These outer zones are connected by an intermediate Prandtl zone, within which the principal stresses rotate continuously through an angle of π . This smooth stress rotation maintains equilibrium for soil condition. The fact that both upper and lower bound solution converge toward this mechanism confirms its significant as the exact solution in limit analysis.

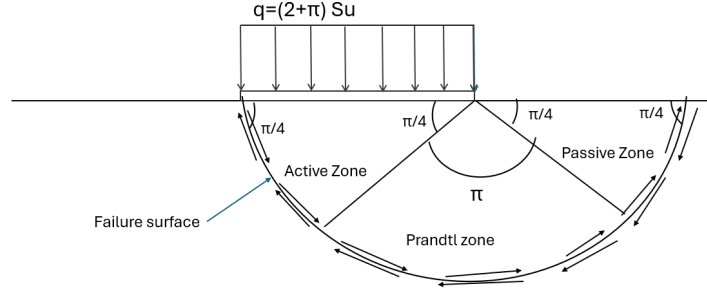


Figure 12: Exact solution for S_u material

3.3 Lower bound finite element limit analysis

Understanding and implementing the lower bound theorem is the is one of the main focus of this project. Lower bound approach is conceptually similar to slip line method, which represents an early form of lower bound analysis, as both methods determine collapse loads based on stress condition (Chen, 1975, p. 107).

The lower bound originates from early slip line field solutions, beginning with Prandtl, 1920 developed the first analytical bearing capacity solution of weightless soil, which was later extended by Reissner (1924) to include embedment effects. Meyerhof in 1938 introduced formulation that account for the soil self weight. subsequently demonstrated that the inclusion of soil self-weight severely limits the applicability of analytical solutions, thereby motivating the use of numerical methods (Chen, 1975, p.1).

Finite element limit analysis integrate the finite element with the limit analysis, by discretising the soil domain into a computational grid. The method enables the numerical reconstruction of complete stress fields and associated failure mechanisms for problems such as bearing capacity and slope stability, showing that realistic collapse state can be obtained through numerical analysis and supporting the development of later finite element formulation Sloan (2013). One of the earliest steps toward finite element analysis was introduced by Lysmer (1970), who incorporated the finite element method into limit analysis, forming the basis of what later became known as finite element analyses framework.

In Lysmer's approach, the soil mass was divided into simple triangular elements, which made it possible to study problems with complex geometry and non uniform soil conditions. By applying the lower bound theorem, collapse loads could be estimated directly, without the need to simulate gradual loading processes Lysmer (1970).

In this formulation, stresses were used as the main unknowns within three noded triangular plane strain element. To make the analysis suitable for the numerical optimization, the nonlinear Mohr Coulomb yield criterion was approximated using a polyhedral surface. This allowed the problem to be solved using linear programming while still satisfying the lower bound condition. Stress discontinuities were permitted between neighboring elements to ensure equilibrium, which helped capture stress variations more realistically. The collapse load was then obtained by finding the maximum load factor that satisfied both equilibrium and yield requirements, resulting in a conservative estimate of the failure load Lysmer (1970).

Despite its solid theoretical basis, early lower bound finite element limit analysis faced practical challenges. At the time, linear programming software took a long time to compute and required the user to assume the shear direction along the potential discontinuities. This limited the methods accuracy.

This difficulties were later reduced by the Sloan (1988), who developed a more efficient active set

optimization algorithms, which improved the method by significantly reducing the computational time and also helped to make two dimensional finite element limit analysis a more practical and accessible tool for geotechnical stability analysis.

A further improvement was proposed by Sloan and Kleeman (1995), who removed the requirement to specify shear directions in advance. Their approach introduced discontinuities along all element boundaries and treated velocity jumps as additional unknowns. This allowed both the position and orientation of shear zones to be identified automatically, leading to more reliable and accurate results. This method also allowed stress discontinuities between element, leading to more accurate results and improved handling of stress singularities Sloan (2013). However, even with this added flexibility, the computational time was still a limitation.

A nonlinear optimization approach to lower bound limit analysis was first developed in Lyamin (1999) and later formulated by Lyamin and Sloan (2002). This work built upon earlier linear programming methods by introducing nonlinear programming techniques and applying them in a three dimensional setting. An important aspect of the approach was that the Mohr Coulomb failure criterion was used in its original nonlinear form, rather than being approximated by a linear yield surface. This allowed soil failure behavior to be presented more realistically while still satisfying the requirement of the lower bound theorem. Although this method was capable of analyzing larger and more complex problems, the increased nonlinearity resulted in higher computational demands.

As curved yield surfaces began to be presented directly, this lead to the development of optimization approaches such as second order cone programming. Makrodimopoulos and Martin (2006) introduced a second order cone programming formulation for lower bound finite element limit analysis of cohesive frictional materials. This optimization method showed that the lower bound limit analysis problem can be expressed directly as a conic optimization problem, allowing the Mohr Coulomb yield criterion to be presented in its mathematical form, rather than through linear approximations Makrodimopoulos and Martin (2006).

Second order cone programming is a convex optimization method designed to efficiently solve problems defined by conic constraints. Since the Mohr Coulomb failure criterion forms a conical surface in stress space, which can be expressed directly using second order cone programming. This allows the yield condition to be enforced exactly while remaining consistent with the lower bound theorem and avoiding yield surface linearisation Makrodimopoulos and Martin (2006).

The second order conic program based formulation provides a practical and efficient way to handle yield conditions that have a conical structure. By doing so, it avoids common numerical difficulties that are often encountered in general nonlinear optimization. This optimization framework can be applied to wide range of problems, although it commonly focuses on the Mohr Coulomb criterion under plane strain conditions. Following work by Krabbenhøft et al., 2008 demonstrated that second order conic programming provides exactly yield representation with efficient and robust solution procedures and computational costs compared to other optimization framework Makrodimopoulos and Martin (2006).

These development forms the basis for the second order conic program based lower bound Finite element limit analysis framework. Second order conic program optimization is computationally efficient and allowed the yield criterion to be enforced in the exact form. For these reasons, the present project focuses on the application of second order conic program to finite element limit analysis.

3.4 Implementation of lower bound finite element limit analysis

In this section, a brief overview of the implementation of lower bound finite element limit analysis method. The theoretical foundations of LB FELA are well established and described in detail by Sloan (1988) and Sloan (2013). Rather than fully derivation, the purpose of this section is to summarise the key concept necessary for understanding the lower bound formulation and its numerical implementation as used in this project.

3.4.1 Finite element discretizations

A three nodes linear stress triangle element is employed, with one node located at each corner. This is the simplest FE commonly used for 2D stress analysis. Within each element stress varies linearly. The stress at node i is defined as:

$$\sigma^i = \begin{bmatrix} \sigma_x^i \\ \sigma_y^i \\ \tau_{xy}^i \end{bmatrix} \quad (4)$$

To ensure a consistent interpolation, the shape functions must satisfy the nodal property:

where σ_x^i and σ_y^i denote the normal stress components, and τ_{xy}^i denotes the shear stress at node i .

The stress field inside the triangular element is interpolated from its nodal shape functions N_i . Shape functions are mathematical interpolation functions that describe how a field variable, such as stress, varies within an element based on its nodal values.

To ensure a consistent interpolation, the shape functions must satisfy the nodal property:

$$N_i = \begin{cases} 1 & \text{for } i = \text{the node} \\ 0 & \text{for } i \neq \text{other node} \end{cases}$$

This condition ensures that the interpolated stress field exactly reproduces the nodal stress value.

For linear triangular element defined in Cartesian coordinates, the shape functions are derived using a linear polynomial basis and can be written in general form as

$$N = N_q A^{-1} \quad (6)$$

$$N = \begin{bmatrix} N_1 \\ N_2 \\ N_3 \end{bmatrix} = \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ x \\ y \end{bmatrix}$$

Where,

$$N = \begin{bmatrix} N_1 \\ N_2 \\ N_3 \end{bmatrix} \text{ is the vector of shape functions}$$

$$N_q = \begin{bmatrix} 1 \\ x \\ y \end{bmatrix} \text{ represents the polynomial basis used for interpolation}$$

$$A = \begin{bmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{bmatrix}^{-1} \text{ is the nodal coordinate matrix defined by the coordinate of the three vertices of the triangular element}$$

For the three nodes triangular element, the shape functions can be expressed as

$$\mathbf{N} = \frac{1}{2A} \begin{bmatrix} x_2 y_3 - x_3 y_2 & y_2 - y_3 & x_3 - x_2 \\ x_3 y_1 - x_1 y_3 & y_3 - y_1 & x_1 - x_3 \\ x_1 y_2 - x_2 y_1 & y_1 - y_2 & x_2 - x_1 \end{bmatrix} \begin{bmatrix} 1 \\ x \\ y \end{bmatrix} \quad (7)$$

where A is the area of the triangle.

The corresponding shape functions are therefor

$$\begin{aligned} N_1 &= \frac{1}{2A} [(x_2y_3 - x_3y_2) + (y_2 - y_3)x + (x_3 - x_2)y] \\ N_2 &= \frac{1}{2A} [(x_3y_1 - x_1y_3) + (y_3 - y_1)x + (x_1 - x_3)y] \\ N_3 &= \frac{1}{2A} [(x_1y_2 - x_2y_1) + (y_1 - y_2)x + (x_2 - x_1)y] \end{aligned} \quad (8)$$

In a lower bound finite element formulation for a plain strain problems, each triangular element carries an independent, varying stress field. The stress at a point within an element is obtained by interpolating the nodal stresses as

$$\sigma = \sum_{i=1}^3 N_i \sigma_i, \quad (9)$$

Consequently, the individual stress components are given by

$$\sigma_x = \sum_{i=1}^3 N_i \sigma_{x_i}, \quad \sigma_y = \sum_{i=1}^3 N_i \sigma_{y_i}, \quad \tau_{xy} = \sum_{i=1}^3 N_i \tau_{xy_i}, \quad (10)$$

3.4.2 Element equilibrium

For a stress field to be statically admissible, the stresses within each triangular element must satisfy the equilibrium equation.

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + b_x = 0, \quad \frac{\partial \sigma_y}{\partial y} + \frac{\partial \tau_{xy}}{\partial x} + b_y = 0 \quad (11)$$

Here,

$$b^e = \begin{bmatrix} b_x \\ b_y \end{bmatrix}$$

denoted the body force vector acting with in element volume. The most common source of body force in geotechnical problem is gravity. Consequently, b_y typically represents the unit weight of the soil, while b_x is often assumed to be zero.

Using the stress interpolation introduced earlier, equilibrium constraints on the nodal stresses can be obtained by substituting the interpolated stress field into the equilibrium equations. This leads to the following element level equilibrium relation:

$$A^e \sigma^e = B^e \sigma^e = b^e \quad (12)$$

B_i are the strain- displacement matrix, which can be defined as,

$$\mathbf{B}^e = \begin{bmatrix} \frac{\partial}{\partial x} & 0 \\ 0 & \frac{\partial}{\partial y} \\ \frac{\partial}{\partial y} & \frac{\partial}{\partial x} \end{bmatrix} \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 \end{bmatrix}$$

which expands to

$$\mathbf{B}^e = \begin{bmatrix} \frac{\partial N_1}{\partial x} & 0 & \frac{\partial N_2}{\partial x} & 0 & \frac{\partial N_3}{\partial x} & 0 \\ 0 & \frac{\partial N_1}{\partial y} & 0 & \frac{\partial N_2}{\partial y} & 0 & \frac{\partial N_3}{\partial y} \\ \frac{\partial N_1}{\partial y} & \frac{\partial N_1}{\partial x} & \frac{\partial N_2}{\partial y} & \frac{\partial N_2}{\partial x} & \frac{\partial N_3}{\partial y} & \frac{\partial N_3}{\partial x} \end{bmatrix}$$

Since the shape function for a linear triangular element vary linearly, their derivatives are constant within the element. The Matrix $\mathbf{B}^e$ can therefore be written as

$$\mathbf{B}^e = \frac{1}{2A^e} \begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & b_1 & 0 & b_2 & 0 & b_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix}$$

where A^e is the area of the triangular element, and the constants b_i and c_i depend on the nodal coordinate Sloan, 2013

Using this formulation, the equilibrium constraints on the nodal stresses for the element e can be written as

$$A^e \boldsymbol{\sigma}^e = -\mathbf{b}^e \quad (13)$$

with

$$A^e = \frac{1}{2A^e} \begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & b_1 & 0 & b_2 & 0 & b_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix}$$

$$[\mathbf{b}^e]^T = [0, -b_y^e]$$

and the vector of nodal stresses for the element e given by

$$\boldsymbol{\sigma}^e = [[\sigma_{x1}]^e \quad [\sigma_{y1}]^e \quad [\tau_{xy1}]^e \quad [\sigma_{x2}]^e \quad [\sigma_{y2}]^e \quad [\tau_{xy2}]^e \quad [\sigma_{x3}]^e \quad [\sigma_{y3}]^e \quad [\tau_{xy3}]^e]$$

3.4.3 Discontinuity equilibrium

Statically admissible stress fields allow discontinuities along the edges between adjacent finite elements. In a lower bound finite element formulation, stresses may be discontinuous across element boundaries, provided that equilibrium conditions are satisfied.

For a system to remain in equilibrium, the forces acting on a discontinuity shared by two adjacent elements must be in balance.

The normal stress σ_n acting on the shared edge must be continuous. If this condition were violated, gaps or overlaps would occur between adjacent elements, which contradicts equilibrium requirements. Similarly, the shear stress τ acting on the shared edge must also be continuous. Otherwise, one element would slide relative to the other, again violating equilibrium.

In contrast, tangential stress components acting parallel to the discontinuity are permitted to be discontinuous. This flexibility is essential in lower bound analysis, as it allows the stress field to approximate the true plastic collapse mechanism more accurately.

3.4.4 Stress transformation

The normal stress σ_n and shear stress τ acting on an inclined plane can be Obtained through stress transformation. Stress transformation enables normal and shear stress to be expressed on a plane of arbitrary orientation, which is essential for evaluating stresses along the element edges and domain boundaries.

The transformation is derived by enforcing equilibrium of a differential soil wedge bounded by the inclined plane and the global Cartesian coordinate. By requiring force equilibrium on this wedge, a relationship between the global Cartesian stress components and the local stress acting on the inclined plane is obtained. A positive clockwise sign convention is adopted.

The stress acting on a plane inclined at an angle θ to the global coordinate system are given by

$$\begin{aligned}\sigma_n &= \sin^2 \theta \sigma_x + \cos^2 \theta \sigma_y + \sin 2\theta \tau_{xy} \\ \tau &= -\frac{1}{2} \sin 2\theta \sigma_x + \frac{1}{2} \sin 2\theta \sigma_y + \cos 2\theta \tau_{xy}\end{aligned}$$

These relations can be written compactly in matrix form using a stress transformation matrix, which maps the Cartesian stress components to the corresponding normal and shear stress acting on an inclined plane.

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} T(\theta) = \begin{bmatrix} \sigma_n \\ \tau_n \end{bmatrix} \quad (15)$$

Where the transformation matrix $T(\theta)$ is defined as

$$T(\theta) = \begin{bmatrix} \sin^2 \theta & \cos^2 \theta & -\sin 2\theta \\ -\frac{1}{2} \sin 2\theta & \frac{1}{2} \sin 2\theta & \cos 2\theta \end{bmatrix}$$

3.4.5 Discontinuity constraints

Two triangular elements who share a common edge, along the edge, equilibrium requires that both elements produce the same normal stress σ_n and shear stress τ . This implies that the stress at nodes 1= 2 and also 3 = 4 must be equal for the two elements.

Since each element is a three node triangle, the stress vary linearly along each elements edges. Consequently, four equality constraints are enforced by requiring continuity of normal and shear stress at the two nodes defining the shared edge:

$$\begin{aligned}\sigma_n^1 &= \sigma_n^2, & \sigma_n^3 &= \sigma_n^4, \\ \tau_n^1 &= \tau_n^2, & \tau_n^3 &= \tau_n^4\end{aligned} \quad (17)$$

These constrains can be summarized in matrix form, which is convenient for numerical implementation and optimization:

$$[A_{\text{equal}}^d] \{\sigma^d\} = \{b_{\text{equal}}^d\} \quad (18)$$

Here, the constraint matrix is given by

$$[A_{\text{equal}}^d] = \begin{bmatrix} T & -T & 0 & 0 \\ 0 & 0 & T & -T \end{bmatrix}$$

where T is the transformation matrix defined previously

The vector of discontinuity stresses is defined as

$$\{\sigma^d\}^T = \{\{\sigma_{x1}, \{\sigma_{y1}, \{\tau_{xy1}, \{\sigma_{x2}, \{\sigma_{y2}, \{\tau_{xy2}, \{\sigma_{x3}, \{\sigma_{y3}, \{\tau_{xy3}, \{\sigma_{x4}, \{\sigma_{y4}, \{\tau_{xy4}\}$$

and the right hand side vector is

$$\{b_{\text{equal}}^d\}^T = \{0 \ 0 \ 0 \ 0\}$$

3.4.6 Stress Boundary conditions

At the boundary of a soil mass subjected to external loading, both normal and shear stress may be present. A normal stress σ_n develops due to the soil weight and externally applied loads acting perpendicular to the boundary, shear loading τ_n may act along the boundary as a result of friction between the soil and the applied loading. These stresses defined the stress boundary conditions acting on the soil mass.

External loading along the boundary is expressed in term of traction, representing force per unit area. The traction vector at node i is given by

$$\mathbf{t}^i = \begin{bmatrix} \sigma_{\mathbf{n}}^i \\ \tau_{\mathbf{n}}^i \end{bmatrix} \quad (19)$$

Within a linear triangular element, stresses vary linearly along the edges. Therefore, boundary conditions are enforced at the nodes defining a boundary edge, in a similar way to the discontinuity constraints:

$$\begin{aligned} \sigma_n^1 &= \sigma_{\mathbf{n}}^1 & \sigma_n^2 &= \sigma_{\mathbf{n}}^2 \\ \tau_n^1 &= \tau_{\mathbf{n}}^1, & \tau_n^2 &= \tau_{\mathbf{n}}^2 \end{aligned} \quad (20)$$

These constraints can be written in a matrix form as

$$[A_{\text{boundl}}^b] \{\sigma^b\} = \{b_{\text{bound}}^b\} \quad (21)$$

where

$$[A_{\text{boundl}}^b] = \begin{bmatrix} T & 0 \\ 0 & T \end{bmatrix} \quad T \text{ is the transformation matrix}$$

The boundary stress vector is defined as

$$\{\sigma^b\}^T = \{\{\sigma_{x1}, \{\sigma_{y1}, \{\tau_{xy1}, \{\sigma_{x2}, \{\sigma_{y2}, \{\tau_{xy2}\}$$

with the prescribed traction vector

$$\{b_{\text{bound}}\}^T = \{\sigma_1 \ \tau_1 \ \sigma_2 \ \tau_2\} \text{traction forces from equation XXX}$$

3.4.7 Yield conditions

The yield criterion everywhere on the domain

$$F(\sigma) \leq 0$$

this requires that the stress state does not violate the soils shear strength.

3.5 Numerical solution: finite element limit analysis

Finite element model OptumG2 can also predict the lower bound and upper bound, thereby bounding the true collapse load.

As the mesh is refined (as the number of elements increase), the numerical solution typically becomes more accurate and convergence towards the exact collapse load. This mesh convergence behavior shown in 13 Krabbenhøft et al. (2016).

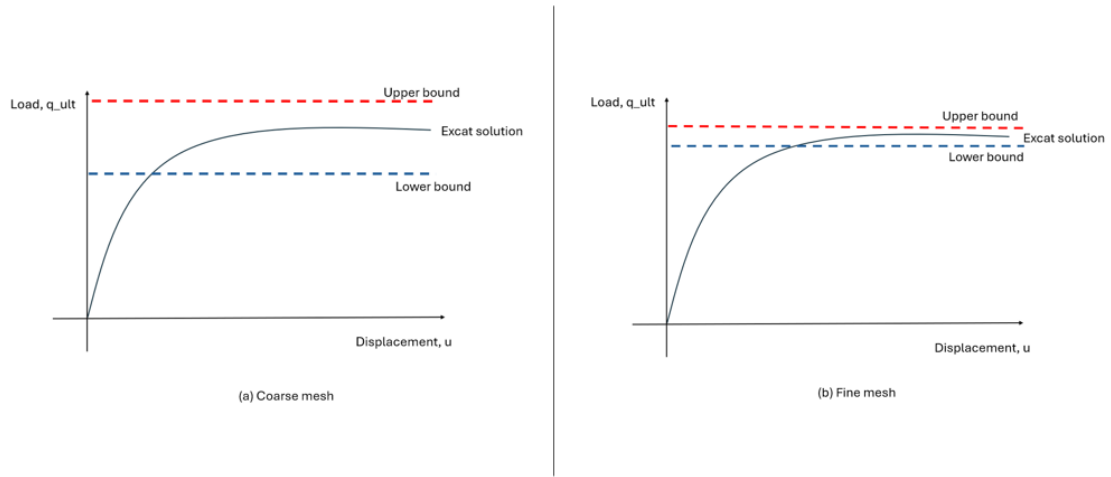


Figure 13: Comparison of coarse and fine finite element meshes used in the bearing capacity analysis

4 Case study: Undrained bearing capacity of a strip footing

Case study on the undrained bearing capacity of a strip footing under vertical load on a clay soil. The objective is to compare bearing capacity estimate obtained from OPTUMG2 with the values from rigid plastic limit load analysis.

4.1 Problem description

The bearing capacity problem shown in figure 14 is analysed using a Tresca soil model, with parameters listed in table 1. The numerical model has a 12 m width and 4 m height. Several mesh densities were considered, both with and without adaptive mesh refinement were considered to assess the effect of meshing on the predicted collapse load. .

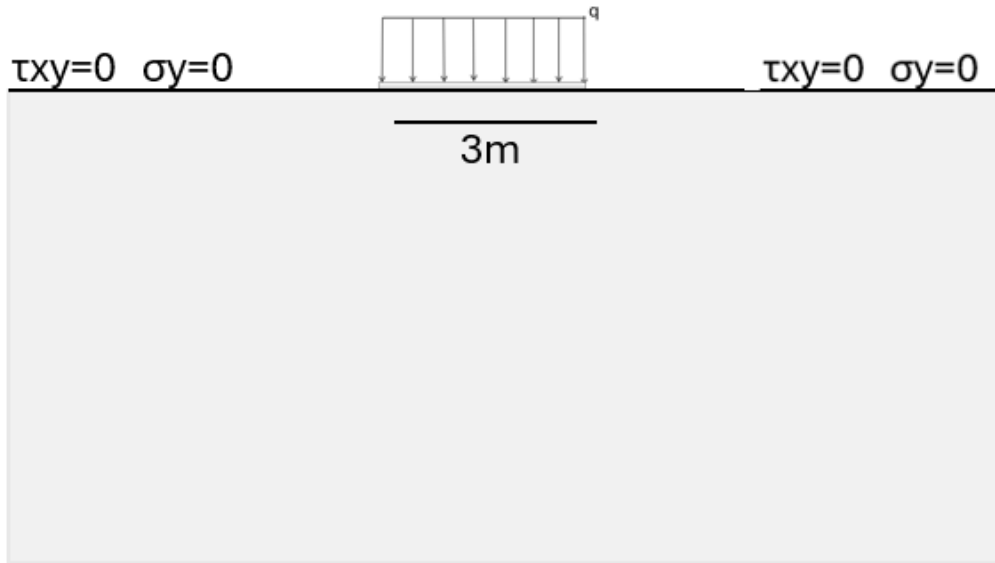


Figure 14: Boundary conditions for the bearing capacity problem

Table 1: Parameters for undrained bearing capacity analysis model

Component	Description	Parameter	Value
Clay - tresca model	Undrained shear strength	S_u	10 kPa
	Unit weight	γ	18 kN/m ³
	Young's modulus	E_u	5000 kPa
	Poisson's ratio	ν	0.49
	Friction angle	ϕ	0°
Footing - Plate	Weight	W	20 kN/m ²
	Width	B	3 m
	Axial stiffness	EA	5×10^7 kN/m
	Bending stiffness	EI	4×10^7 kN m ² /m

4.1.1 Exact analytical solution

For an undrained Tresca material, the exact bearing capacity of a strip footing is given by

$$q_{\text{exact}} = (2 + \pi) S_u + P$$

where $N_c = 2 + \pi$ is the bearing capacity factor, S_u is the undrained shear strength, and P is the overburden pressure.

For this case the the the, since there was no overburden pressure, we can find the ultimate bearing capacity by

$$\begin{aligned} q_{\text{exact}} &= 5.14 \tau_c \\ \tau_c = \tau_{\text{failure}} &= S_u = 10kPa \end{aligned}$$

$$q_{\text{exact}} = 51.4kPa$$

The analytical solution provides that the exact collapse load is $q_{\text{exact}} = 51.4kPa$.

4.2 Numerical result from OptumG2

The bearing capacity problem was computed using OptumG2, which employs SOCP optimization to solve the limit analysis. The analysis was performed by computing the lower bound and upper bound collapse loads. The true collapse load lie between these two solution.

$$\text{Lower bound collapse load} \leq \text{True collapse load} \leq \text{Upper bound collapse load}$$

Since the analytical solution for the solution for the problem is $q_{\text{exact}} = 51.4kPa$, the numerical estimates are expected to converge toward this value with increasing mesh refinement.

4.2.1 Model geometry and meshing

The model geometry and element discretisations used in OptumG2 analyses are shown in figure 15. Both uniform and adaptive mesh refinement strategies were considered to investigate their influence on the estimated collapse load.

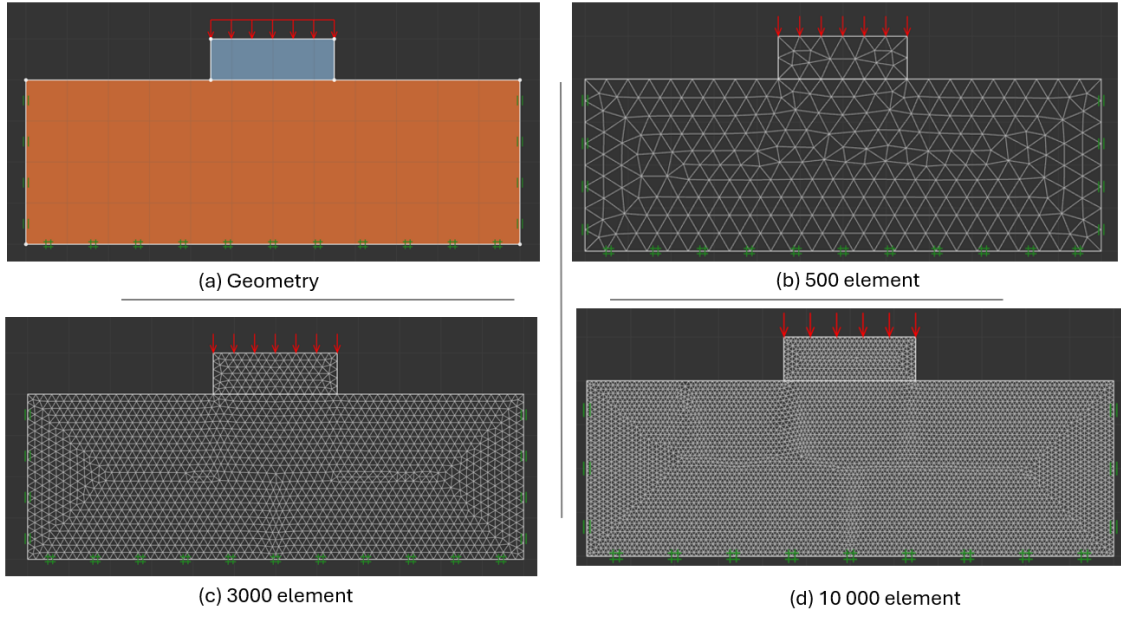


Figure 15: Geometry and meshing used in OptumG2 for the undrained bearing capacity limit analysis

4.2.2 Uniform mesh refinement

Result

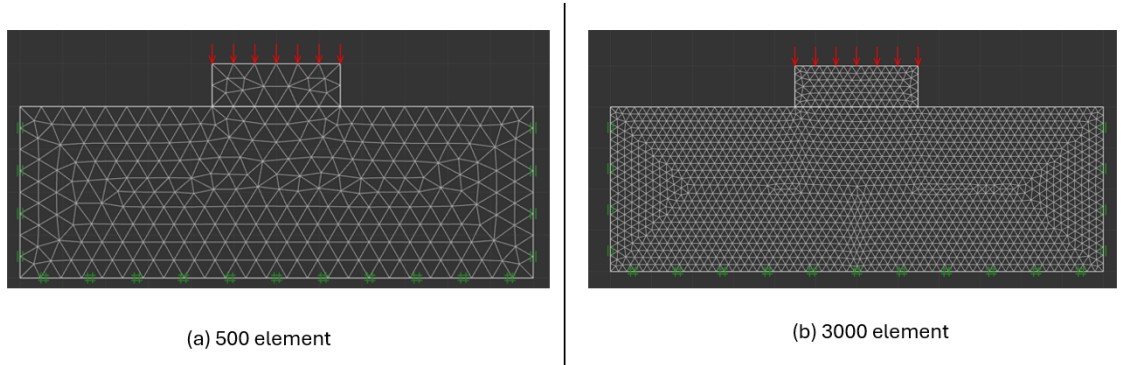


Figure 16: Bearing capacity limit analysis using three iteration of uniform meshing refinement controlled by shear dissipation

Table 2: Lower bound and upper bound element ultimate load estimates from OptumG2 model for two uniform mesh densities,

Element	LB [kPa]	UB [kPa]
500	47,806	57,827
3000	49,965	54,123

Discussion

The result obtained using uniform mesh refinement show a clear convergence trend as mesh is refined. The gap between the LB and UP decreases with increasing mesh density, indicating improved resolution of the failure mechanism.

The relative coarse meshes gives a estimate collapse load remains within reasonable proximity of the theoretical solution.

4.2.3 Adaptive mesh refinement

Result

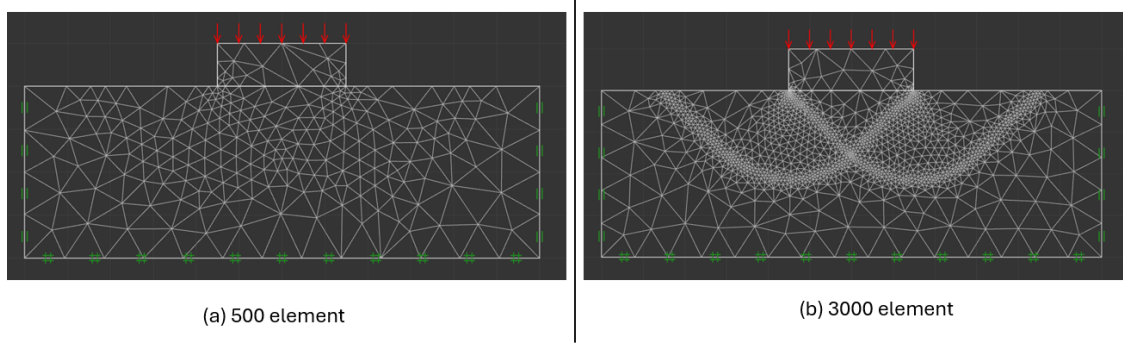


Figure 17: Bearing capacity limit analysis using three iteration of adaptive meshing refinement controlled by shear dissipation

Table 3: Lower bound, upper bound ultimate load estimates from OptumG2 model for two adaptive mesh refined densities

Element	LB [kPa]	UB [kPa]
500	48,345	54,574
3000	50,615	52,216

Discussion

The adaptive mesh refinement result shown in Figure 16 and table 2 demonstrate rapid and efficient convergence toward the analytical collapse load.

The LB and UB solutions become closely spaced relatively low global mesh densities, indicating that the governing plastic mechanism is being resolved accurately. This behavior is a direct consequence of the adaptive mesh refinement strategy, which concentrates mesh refinement in regions of higher shear dissipation where plastic deformation localities. By refining only these critical zones, the failures mechanism is captured in a greater detail without unnecessary refinement elsewhere in the domain.

In contrast, uniform mesh refinement adds elements throughout the domain, leading to convergence towards the analytical solution but requiring significantly more element to achieve similar accuracy. The lower - upper gap therefore closes more slowly, indicating lower computational efficiency. A comparison between tabel 2 and 3 shows that adaptive refinement attains comparable or better accuracy with fewer elements, highlighting its effectiveness for bearing capacity problems governed by localized plasticity.

4.2.4 adaptive mesh with 10 000 elements

Result

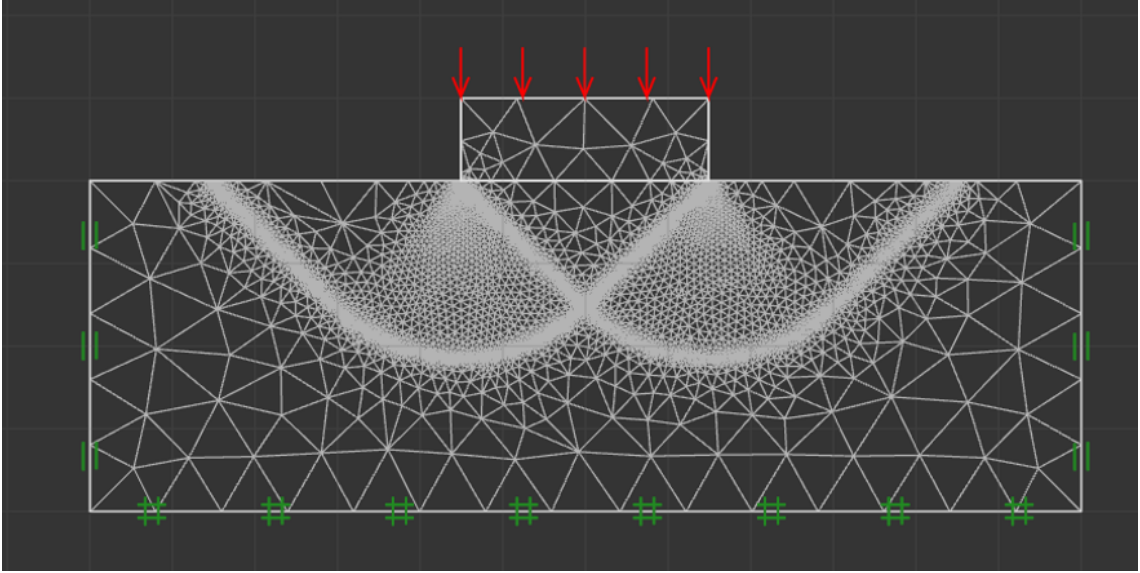


Figure 18: Bearing capacity limit analysis using three iteration of adaptive meshing refinement controlled by shear dissipation for 10 000 element, computes a result that is very close to the theoretical collapse load

Table 4: Lower bound and upper bound ultimate load estimates from OptumG2 model for 10 000 element with adaptive mesh refined

Element	LB [kPa]	UB [kPa]
10 000	51,046	51,791

Discussion The highest resolution analysis was performed using an adaptive mesh consisting of 10,000 elements. The result in figure 18 and table 4 produces LB and UB load that converges to nearly the same value, indicating that the failure mechanism is almost fully resolved. The very small remaining gap between the bounds suggest that numerical solution is extremely close to the exact collapse load for the Tresca material. Compared with the previous adaptive and uniform mesh analyses, the high resolution adaptive mesh provides the most accurate and robust estimate, confirming the effectiveness of targeted mesh refinement in limit analysis.

4.2.5 Comparison of numerical results with theoretical solution

Figure 19 illustrate that all numerical approaches reproduce the same Prandtl type failure mechanism predicted by the analytical solution. The adaptive OptumG2 result shows a more sharply localized collapse zone due to targeted refinement.

Overall, the comparison confirms that despite difference in formulation and meshing strategy, all methods converge to the same collapse mechanism when sufficient refinement is employed.

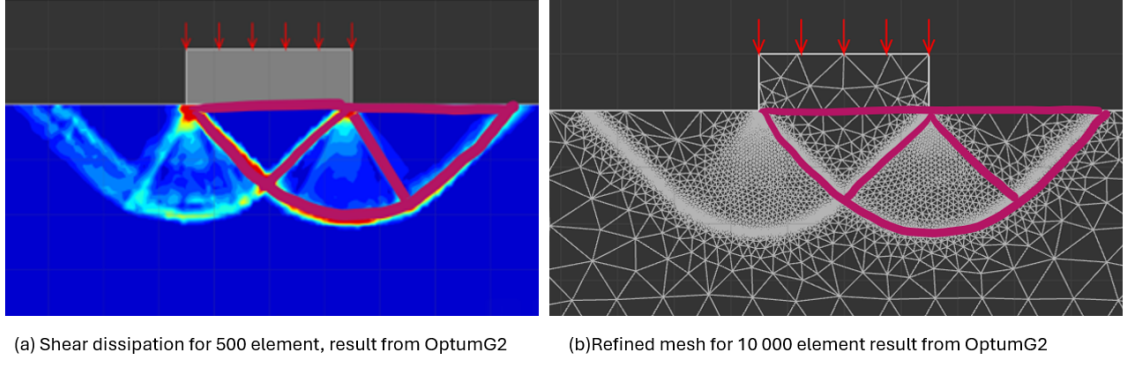


Figure 19: Comparison of the result obtained from OPTUMG2 with Prandtl failure mechanism

4.3 Summary of case study findings

This case study demonstrated that OPTUMG2 model provide upper and lower collapse load estimate that converges to the exact solution, is able able to reproduce the classical Prandtl type failure.

Table 5: Lower and upper bounds of q for uniform and adaptive mesh refinement

Mesh refinement strategy	Elements	LB [kPa]	UB [kPa]
Uniform mesh refinement	500	47.806	57.827
	3000	49.965	54.123
Adaptive mesh refinement	500	48.345	54.574
	3000	50.615	52.216
	10 000	51.046	51.791

OptumG2 is based on limit analysis and explicitly computes both LB and UB solutions. These bounds do not guarantee higher accuracy, but they provide a rigorous envelope within which the true collapse load must lie. The availability of this range offers a clear measure of solution robustness and allows the analyst to assess conservatism directly. This is particularly valuable in geotechnical design, where understanding whether a numerical solution predication is conservative or not safe is often as important as the magnitude of the predicted load.

The adaptive mesh strategy in OptumG2 was shown to resolve the failure mechanism more efficiently than uniform refinement by concentrating elements in zones of plastic localization. As a result, adaptive mesh refinement leads to a faster reduction of the LB and UB gap and more accurate identification of the collapse load compared with uniform refinement. This targeted approach allows computational effort to be focused where it is most influential, while also highlighting the mesh dependence of the of the predicted collapse load.

5 Conclusion

This project investigated the undrained bearing capacity of a strip footing using limit analysis. The main objective was to assess whether modern numerical limit analysis, implemented through OPTUMG2, can accurately reproduce the exact solution and evaluate the influence of mesh refinement strategies on solution accuracy and robustness.

The study demonstrated that both lower bound and upper bound solutions obtained from OPTUMG2 converge towards the exact solution obtained analytically for an undrained Tresca material. The analytical collapse load $q_{\text{exact}} = 5.14S_u = 51.4 \text{ kPa}$, was successfully reproduced within a very close bound as mesh refinement increased. This confirms that the objectives of validating the numerical limit analysis against an exact solution were achieved.

A key finding of this project is that adaptive mesh refinement significantly outperforms uniform mesh refinement. By concentrating elements in zones of high shear dissipation, the adaptive refinement resolves the failure mechanism more efficiently and leads to much faster reduction of the gap between LB and UP solutions. With an adaptive mesh of 10,000 elements, the difference between the bounds became extremely small, indicating that the numerical solution was extremely close to the true collapse load.

Unlike conventional displacement based FEM approaches such as PLAXIS, OPTUMG2 explicitly provides both lower and upper bound solution, offering a rigorous envelope within which the true collapse load must lie. This feature allows the conservatism and reliability of the solution to be assessed directly, which is particularly valuable in geotechnical design where safety consideration are critical. Overall, the project confirms that FELA combination with SOCP optimization, is a powerful, efficient and reliable tool for stability analysis problem governed by plastic collapse.

Limitation

Despite the strong agreement between analytical and numerical results, this limitations should be acknowledged.

First, the case study was restricted to a simple geometry and homogeneous soil conditions, assuming plane strain and undrained Tresca behavior. Real geotechnical problems often involve layered soils, anisotropy, groundwater effects, and complex boundary conditions, which were not considered here.

Additionally, although adaptive mesh refinement greatly improved efficiency, the solutions remains mesh dependent, and achieving very close bound requires a sufficiently fine mesh. This can increase computational cost for large scale or three dimensional problems.

Future work

Future research could be extended this work in several important directions. First, further development and investigation of SOCP based optimization for LB FELA would be valuable, as SOCP allow Mohr Coulomb yield criterion to be presented exactly rather than approximated, improving both numerical accuracy and computational efficiency.

Additional, future research could focus on extending the approach to a diverse stability problems and further improving adaptive mesh strategies, for example by developing more advanced refinement criteria, combining adaptive meshing with fan type elements to better capture stress singularities.

Finally, another possible direction for future work is the integration of machine learning techniques into adaptive mesh refinement strategies. Data driven models could be trained to identify regions where plastic localization is likely to occur, allowing the mesh to be refined more systematically and efficiently. Such an approach has the potential to significantly reduce computational time and cost while maintaining accuracy.

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